

Cyber Security PSBs Hackathon Series – 2026. An Initiative by DFS & IBA via PSBs, hosted at MNNIT Allahabad, Prayagraj.

Note: Please do not modify the template. The proposal must be limited to a maximum of 3 pages. The completed document should be saved in PDF format and uploaded on the registration-cum-concept note form.

1. Team Name:

ZeroIntent

2. Title of the Proposed Idea:

PhantomGrid – A Three-Layer Passive Behavioural Authentication System for PSB Digital Channels

(Concise, descriptive, and solution-focused)

3. Problem Statement

- **Cybersecurity problem being addressed:** Account Takeover (ATO) attacks on Public Sector Bank (PSB) digital channels remain critically under-addressed at the session level. Stolen credentials, session hijacking, and SIM-swap attacks allow adversaries to authenticate successfully and operate undetected because no continuous re-verification of the legitimate user occurs after login. Existing controls — OTPs, PINs, passwords — are point-in-time gates that an attacker clears once and exploits indefinitely.
- **Relevance to public sector banking:** PSBs serve over 600 million account holders across internet banking, mobile banking, and UPI platforms. UPI alone processes over 18 billion transactions monthly in India. A single compromised session can drain savings, reverse months of salary, or expose government subsidy accounts. The scale and the trust deficit that follows each ATO incident makes this a national financial security priority.
- **Key gaps in existing systems:** Current authentication frameworks are entirely credential-centric. They do not model how a legitimate user behaves once inside a session. No existing PSB deployment correlates navigation cognition, transaction intent patterns, and authorization rhythms simultaneously. An attacker with valid credentials faces zero behavioral resistance after the login gate.

4. Proposed Solution (Abstract):

PhantomGrid is a passive, zero-friction, three-layer behavioral authentication mesh that wraps invisibly around every banking session. It requires no additional hardware, no user action, and no change to existing UI. The system continuously verifies that the person operating the session is the same person who enrolled — not through what they know or what they carry, but through how they think, how they navigate, and how their fingers move.

Layer 1 — CognitiveTrap (Session Navigation): A dynamically generated invisible canvas overlay renders personalized decoy elements in zones the legitimate user has never historically interacted with. A genuine user on motor-autopilot never touches these zones. An attacker reading the interface for the first time will hover, hesitate, or interact with decoys — triggering an immediate behavioral flag. Simultaneously, cognitive latency on familiar UI zones is measured; attackers reading a new interface show 200–400ms hesitation spikes that genuine users never exhibit.

Layer 2 — IntentTrace (Transaction Approach): As the user approaches a UPI transaction, three independent signals are captured: (a) navigation entropy — a legitimate user follows a low-entropy habitual path to payment; an attacker explores and backtracks; (b) digit entry fluency on the amount field — familiar amounts are typed in one fluid burst while unfamiliar amounts show inter-digit micro-hesitations of 180–300ms; (c) beneficiary hover dwell time — a recognized contact is confirmed in under 700ms while an attacker reading an unfamiliar name lingers for 1.8–2.5 seconds.

Layer 3 — RhythmLock (Authorization): The user's 6-digit UPI PIN carries a personal inter-key interval rhythm formed over hundreds of repetitions. PhantomGrid captures this 5-dimensional timing vector and compares it to the enrolled baseline using Dynamic Time Warping. An attacker who knows the correct PIN will enter it with their own rhythm, not the account holder's, and will be caught at the final gate before funds move.

Innovation: No existing PSB solution applies behavioral authentication across all three lifecycle phases simultaneously. PhantomGrid is India-specific in its UPI-native design, undetectable at the DOM level, and computationally lightweight enough for real-time inference on commodity infrastructure.

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5. PI Category:

Software-Centric (with Cloud Integration) — PhantomGrid is a pure software solution. The client-side behavioral event capture module runs as JavaScript within the banking portal. The ML inference engine and risk fusion layer run on the bank's existing server infrastructure (on-premise or private cloud). No specialized hardware is required. The architecture is cloud-ready for horizontal scaling and can be deployed as a microservice alongside existing core banking API gateways.

6. Technical Approach & Methodology:

- **Data Type and Sources:** All data is passively captured from the user's own browser session — no biometric device, no external data feed. Signals include: JavaScript keydown/keyup event timestamps (keystroke dynamics), mousemove and mouseenter event coordinates and durations (hover dwell), browser navigation timing API events (page transition latency), custom canvas interaction events for Layer 1 decoy detection, and sessionStorage-based navigation sequence logs. No Personally Identifiable Information (PII) is extracted — only anonymized behavioral vectors.
- **Technology Stack:**
 - Frontend: Vanilla JavaScript (ES6+) event capture module injected into the banking portal. HTML5 Canvas API for invisible decoy overlay rendering. WebSocket client for real-time signal streaming to backend.
 - Backend: Python 3.11, FastAPI for low-latency REST/WebSocket endpoints. scikit-learn Isolation Forest for Layer 1 and Layer 2 anomaly scoring. Custom Dynamic Time Warping (DTW) implementation for Layer 3 PIN rhythm comparison.
 - Data Storage: PostgreSQL for per-user behavioral baseline profiles. Redis for live session state and streaming risk score cache.
 - Dashboard: React.js analyst console with real-time risk gauge, layer signal breakdown, and alert log.
 - Infrastructure: Docker-containerised microservice, deployable on any cloud or on-premise Linux server.
- **Methodology and Implementation Flow:**
 - Step 1 — Enrollment Phase:** During the first 3–5 legitimate sessions post-onboarding, PhantomGrid silently collects behavioral events and trains per-user Isolation Forest models for Layers 1 and 2. The Layer 3 PIN rhythm baseline is built from the first 5 PIN entry events and stored as an encrypted inter-key interval vector.
 - Step 2 — Live Session Monitoring:** On every subsequent login, the CognitiveTrap overlay activates within 200ms of dashboard render. Events are streamed via WebSocket to the inference backend at 50ms intervals. Each layer's anomaly score is computed independently and fed to the fusion engine.
 - Step 3 — Risk Score Fusion:** The weighted fusion engine (Layer 1: 30%, Layer 2: 40%, Layer 3: 30%) produces a live composite risk score (0–100). Score below 60 = Green (session continues). Score 60–84 = Amber (silent OTP re-authentication overlay, no session disruption). Score 85+ = Red (transaction blocked, session terminated, analyst alerted with full signal breakdown).
 - Step 4 — Continuous Adaptation:** Baseline models update incrementally on confirmed-legitimate sessions using an exponential moving average approach, ensuring that gradual natural behavioral drift (typing speed changes, device changes) does not cause false positives over time.
- **Prototype Scope (4–5 days):** Simulated banking portal with full UPI transfer flow, all three behavioral capture layers active, Isolation Forest + DTW models trained on scripted legitimate vs. attacker sessions, live risk gauge dashboard, and a scripted two-scenario demo (legitimate user: all green; attacker with valid credentials: escalating flags, transaction blocked at Layer 3).

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7. Expected Outcome:

- **Results delivered:** PhantomGrid will detect illegitimate session takeovers with high accuracy across three independent behavioral signals. The system will block unauthorized transactions before funds leave the account, generate explainable per-signal risk alerts for security analysts, and operate with sub-15ms inference latency per scoring cycle — fully transparent to the end user.
- **Performance targets for prototype:**
 - Layer 1 (CognitiveTrap): Decoy interaction detection rate $\geq 95\%$ on scripted attacker sessions; false positive rate on legitimate sessions $\leq 2\%$.
 - Layer 2 (IntentTrace): Isolation Forest anomaly detection on attacker navigation + amount entry $\geq 88\%$ accuracy on held-out test sessions.
 - Layer 3 (RhythmLock): DTW distance-based PIN rhythm mismatch detection $\geq 92\%$ accuracy; false rejection rate on legitimate user $\leq 3\%$.
 - Composite system (all three layers): End-to-end ATO session blocking rate $\geq 96\%$ on attacker sessions before transaction authorization is reached.
- **Risk score latency:** $\leq 15\text{ms}$ per scoring cycle on a standard 4-core CPU server, making the system viable for real-time session monitoring at scale without adding perceptible latency to the user's banking experience.

8. Impact & Feasibility:

- **Impact on banks:** PSBs gain a continuous session authentication layer that operates entirely within existing digital infrastructure — no hardware procurement, no UI redesign, no customer enrollment friction. The analyst dashboard provides explainable, auditable risk signals that meet RBI's cyber security framework requirements for behavioral monitoring. Estimated reduction in ATO-related fraud losses: 60–80% for sessions reaching the payment stage.
- **Impact on customers:** Legitimate users experience zero friction — no additional OTPs, no CAPTCHAs, no behavioral surveys. The system is entirely invisible during normal use. Customers benefit only when an attacker is caught, and the catch happens before money moves.
- **Scalability and real-world applicability:** PhantomGrid's client module is a JavaScript snippet injectable into any existing web banking portal without framework changes. The backend microservice scales horizontally behind a load balancer. The UPI-native design is directly applicable to the NPCI ecosystem and to all 12 major PSBs operating UPI-enabled digital channels. Mobile banking extension is achievable via a React Native or Android SDK implementation of the same capture module.
- **Practical feasibility:** All components use open-source, production-proven libraries (scikit-learn, FastAPI, PostgreSQL, Redis). There are no proprietary API dependencies, no third-party biometric data processors, and no regulatory blockers to deployment within a PSB's existing technology stack. A functional prototype demonstrating all three layers is achievable within 4–5 days using scripted session data.
- **Challenges and limitations:** Cold-start period of 3–5 sessions is required before per-user models become reliable — new accounts are protected at a reduced sensitivity level during enrollment. Users with severe motor impairments or highly inconsistent device usage patterns may require longer enrollment periods or adjusted anomaly thresholds. Layer 3 PIN rhythm detection is currently designed for web-based PIN entry; native mobile PIN pads with hardware masking will require a platform-specific adaptation. These are known, solvable engineering challenges, not fundamental design limitations.